Supplement 3: The Regularized Horseshoe for Covariate Selection Improves Convenience and Predictive Performance in Population PK/PD Models

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1 Simulation Details

The simulated data example consisted of a standard two-compartment population PK model with IV dosing. All subjects were given the same dose of 400mg infused over one hour. PK observations were made subsequently at hours 1, 2, 4, 6, 24, 48, 72, 168, and 336 for all subjects. Observational noise was assumed to be log-normal with $\sigma = 0.2$. The true values of population parameters were set to the values shown in Table S1 and correspond to a typical terminal half-life of 10.4 days.

To add more covariates to the dataset, eGFR, albumin, and disease severity values were also simulated. These covariates were simulated as independent from each other and to the other seven covariates gathered from the NHANES dataset. To simulate realistic values of eGFR, subjects were randomly selected to belong to one of four eGFR categories (severe loss of kidney function, moderate loss of kidney function, mild loss of kidney function, and normal kidney function) with probabilities 0.1, 0.2, 0.3, and 0.4, respectively. Conditional on the category selected, a value was then drawn uniformly from the intervals (15,30), (31,60), (61,90), and (91,120) depending on the category and then rounded to the nearest integer. Albumin and disease severity were simulated more readily by simply drawing from a $LogNormal(\ln(4.5), 0.3)$ and $LogNormal(ln(40), 0.5)$, respectively.

Ground-truth values of effect sizes of 1, 0.5, 0.25, and 0.05 were used to get a sense how each method did across a range of varying realistic effect sizes. Ground truth values of the parameter-covariate parameters for each sparsity setting are presented in Table S2.

Table S1: Ground-truth values of population parameters used for simulation study.

Parameter	**Typical Value**	ω
CL	0.02 L/h	0.3
$V1$	3 L	0.3
Q	0.2 L/h	0.3
$V2$	4 L	0.3

Table S2: Ground-truth values of parameter-covariate effects at each sparsity level.

Sparsity	Par	AGE	SEX	WT	HT	BMI	LBM	BSA	EFGR	ALB	DIS
0	CL	0	0	0	0	0	0	0	0	0	0
0	$V1$	0	0	0	0	0	0	0	0	0	0
0	Q	0	0	0	0	0	0	0	0	0	0
0	$V2$	0	0	0	0	0	0	0	0	0	0
0.1	CL	1	0	0	0	0	0	0	0	0	0
0.1	$V1$	0.5	0	0	0	0	0	0	0	0	0
0.1	Q	0.25	0	0	0	0	0	0	0	0	0
0.1	$V2$	0.05	0	0	0	0	0	0	0	0	0
0.5	CL	1	0.25	0.05	0	0	0	0	0	1	0.5
0.5	$V1$	0.5	0.05	1	0	0	0	0	0	0.5	0.25
0.5	Q	0.25	1	0.5	0	0	0	0	0	0.25	0.05
0.5	$V2$	0.05	0.5	0.25	0	0	0	0	0	0.05	1
0.7	CL	1	0.25	0.05	0	0	0	0.25	0.05	1	0.5
0.7	$V1$	0.5	0.05	1	0	0	0	0.05	1	0.5	0.25
0.7	Q	0.25	1	0.5	0	0	0	1	0.5	0.25	0.05
0.7	$V2$	0.05	0.5	0.25	0	0	0	0.5	0.25	0.05	1

2 Simulated Data Results

2.1 Evaluation Metrics

Figures S1, S2, and S3 show the evaluation metrics of the estimators across the various sparsity values and dataset sizes for the age covariate against $V1$, Q, and $V2$ respectively. The respective ground truth values are 0.5, 0.25, and 0.05 to illustrate how properties of the method may change depending on the effect size.

Figure S1: Evaluation metrics of estimators of $\theta_{V1,Age}$ as a function of ground-truth sparsity value and number of subjects fit. Nonzero ground-truth value is 0.5.

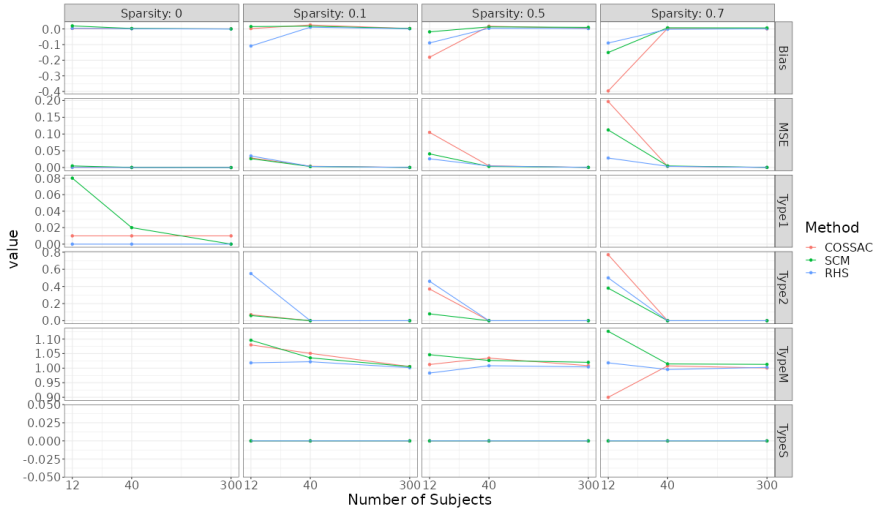


Figure S2: Evaluation metrics of estimators of $\theta_{Q,Age}$ as a function of ground-truth sparsity value and number of subjects fit. Nonzero ground-truth value is 0.25.

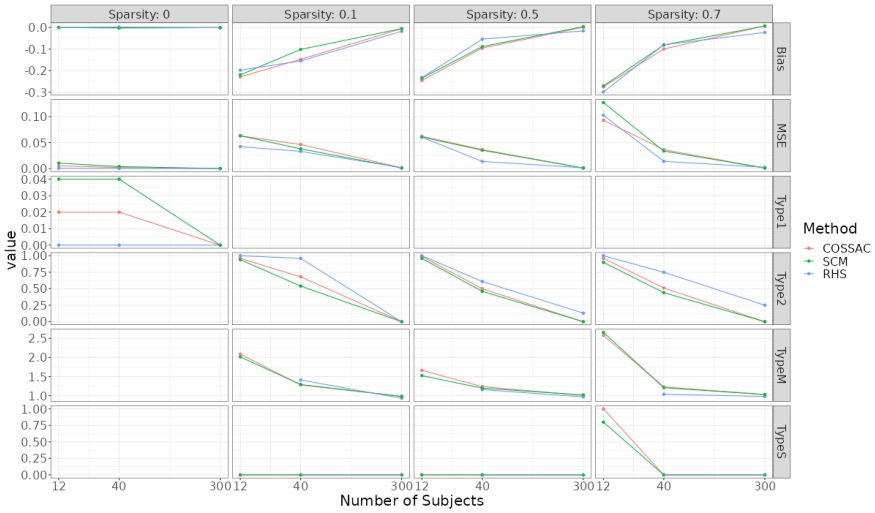
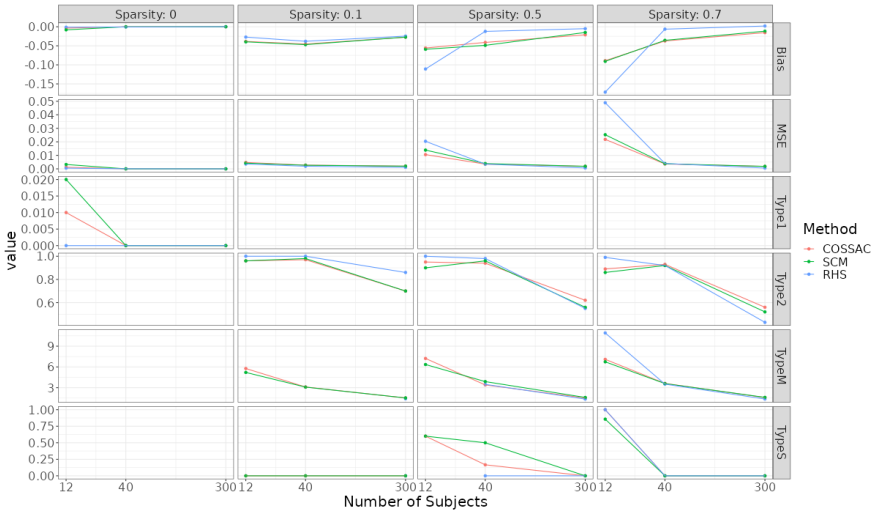


Figure S3: Evaluation metrics of estimators of $\theta_{V2,Age}$ as a function of ground-truth sparsity value and number of subjects fit. Nonzero ground-truth value is 0.05.



Notably, compared to the CL-Age coefficient shown in the main paper which shows a clear upward bias, for these coefficients with smaller true effect size, the bias for the stepwise methods is often negative. This is because for these smaller effect sizes the stepwise methods are more likely to estimate a value

of zero, bringing down the estimate on average. The TypeM error shows that when the stepwise methods do declare these effects significant, they still tend to exaggerate their effect sizes. In addition, the TypeM error seems to be larger when the true effect size is smaller.

Also of note is that for the smaller true effect sizes and less sparsity, the TypeS error seems to grow, especially in the stepwise methods. In other words, when they do declare these effects significant they estimate the effects with a sign that is the opposite of the truth more often, leading to misleading or potentially harmful inferences.

In addition, we also examine the metrics on another covariate, weight, that is much more correlated with other covariates to examine how correlation affects the performance of the methods. Figure S4 shows the evaluation metrics for the weight covariate against the $V1$ parameter which has a true effect size of one. Notably, for $\theta_{V1,Weight}$ the RHS point estimates tend to have a large downward bias and a correspondingly higher Type2 error rate. This is not surprising, however. Since weight is highly correlated with several other covariates and effects are estimated jointly, the effect can be "spread out" over the covariates especially when averaged over all posterior samples.

Figure S4: Evaluation metrics of estimators of $\theta_{V1,Weight}$ as a function of ground-truth sparsity value and number of subjects fit. Nonzero ground-truth value is 1.

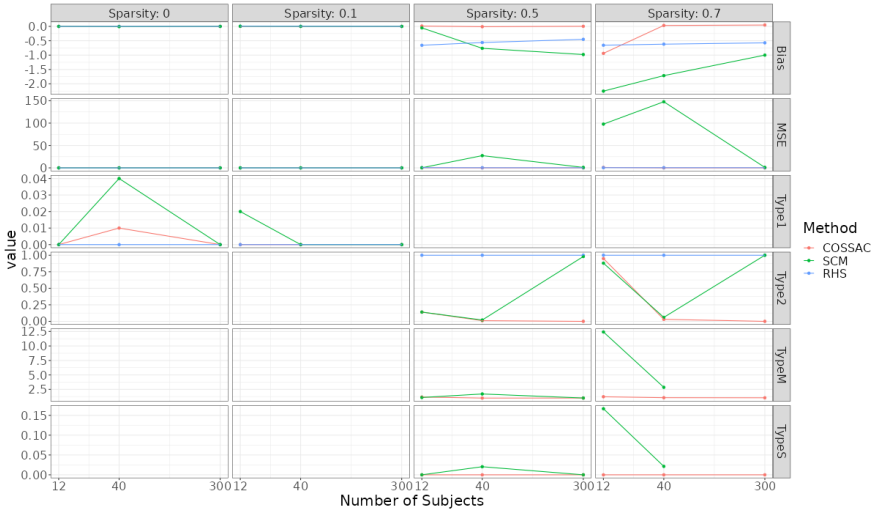
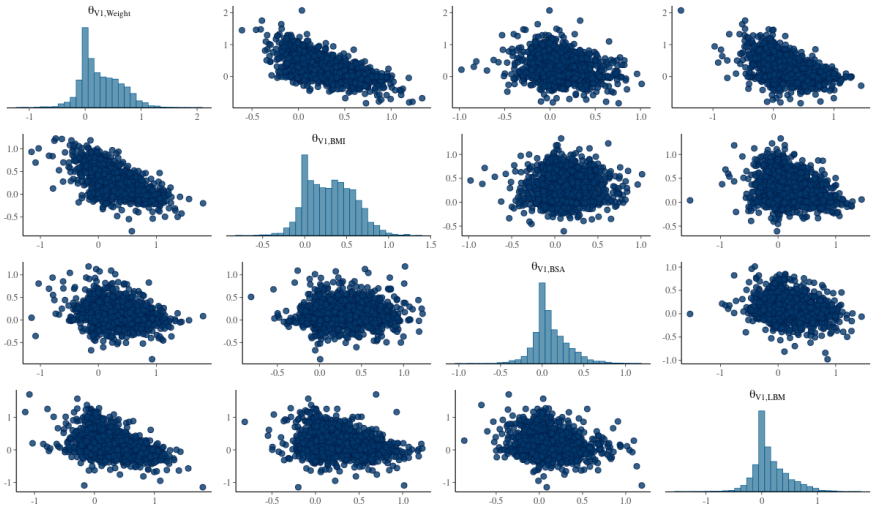


Figure S5 shows posterior samples of $\theta_{V1,Weight}$ along with the three other highly correlated covariate effects on weight. While the marginal posteriors tend to show concentration at zero due to the sparsity effects of the RHS, most of the samples seem to be non-zero for more than one covariate, allowing for the "spreading out" of the effect over the correlated covariates. Meanwhile the stepwise approaches are able to better select weight as the true covariate, but

when they do they still show a high TypeM error, possibly giving a trade-off of sparsity in exchange for worse predictive power.

Figure S5: Scatterplots of MCMC posterior samples for $\theta_{V1,Weight}$, $\theta_{V1,BMI}$, $\theta_{V1,BSA}$, $\theta_{V1,LBM}$.



Figures S6, S7, and S8 show the metrics for the effect of weight on the other parameters with smaller true effect sizes. Since the true effect sizes are smaller, even the stepwise approaches usually fail to declare these parameters significant.

Figure S6: Evaluation metrics of estimators of $\theta_{Q,Weight}$ as a function of ground-truth sparsity value and number of subjects fit. Nonzero ground-truth value is 0.5.

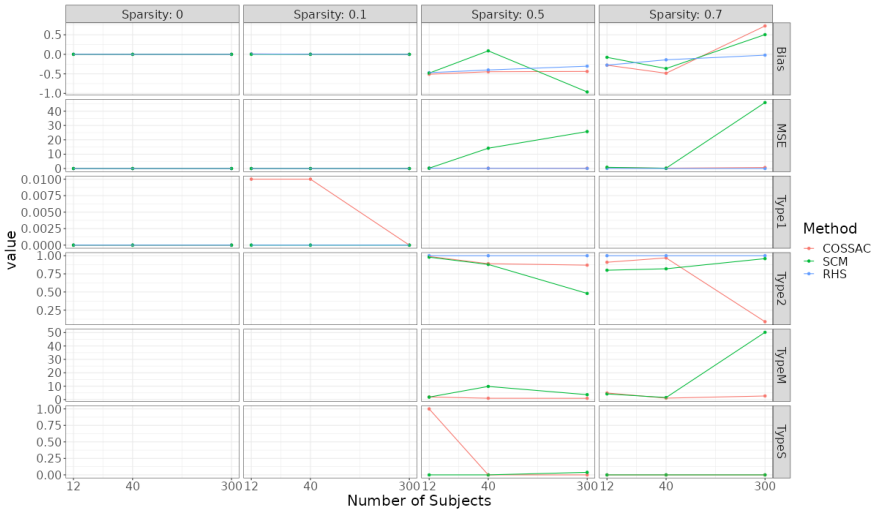


Figure S7: Evaluation metrics of estimators of $\theta_{V2,Weight}$ as a function of ground-truth sparsity value and number of subjects fit. Nonzero ground-truth value is 0.25.

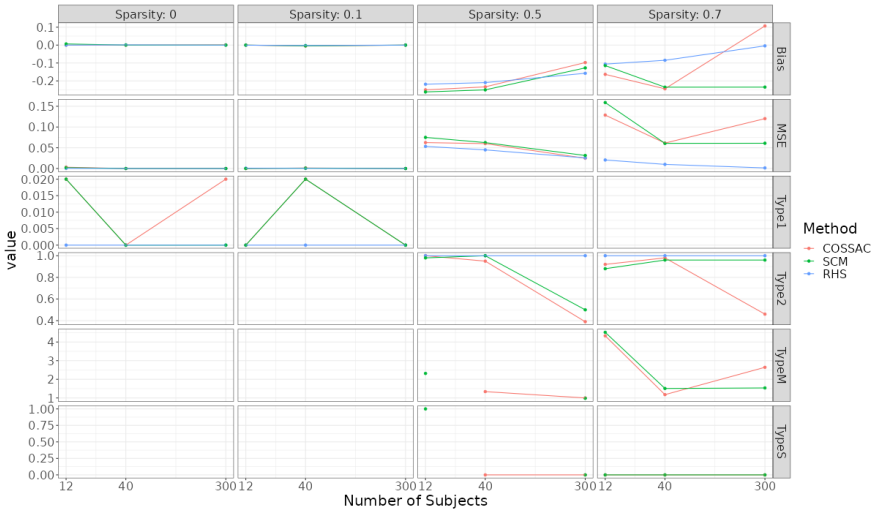
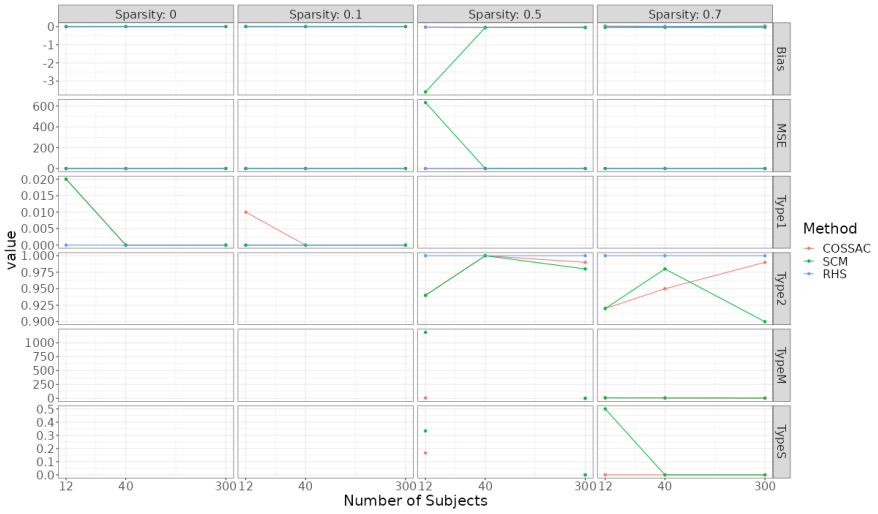


Figure S8: Evaluation metrics of estimators of $\theta_{CL,Weight}$ as a function of ground-truth sparsity value and number of subjects fit. Nonzero ground-truth value is 0.05.



2.2 LOGO-CV

Figure S9 shows the LPPD for each method and each simulation setting averaged across the 100 replicated of the dataset. RHS shows superior LPPD at all settings, but particularly for lower numbers of subjects.

Figure S9: Average LPPD for each method at all simulation settings.

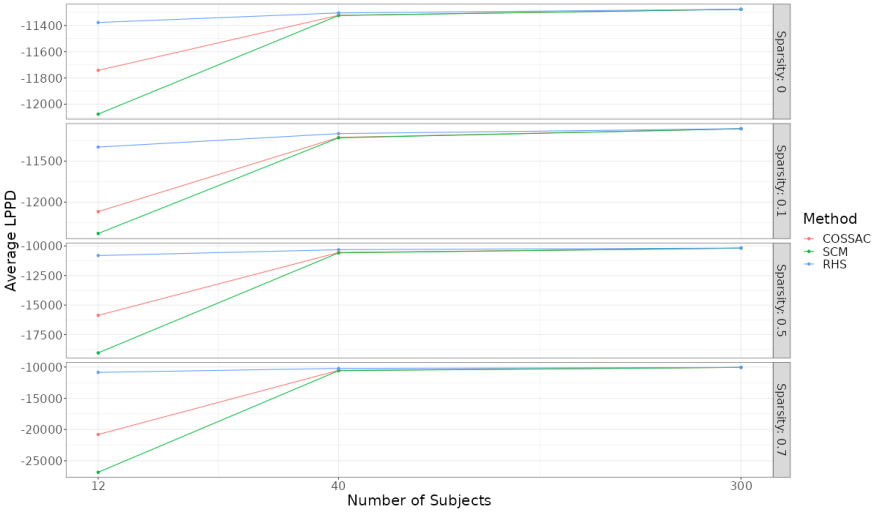


Figure S10: LPPD of each method for each simulation setting.

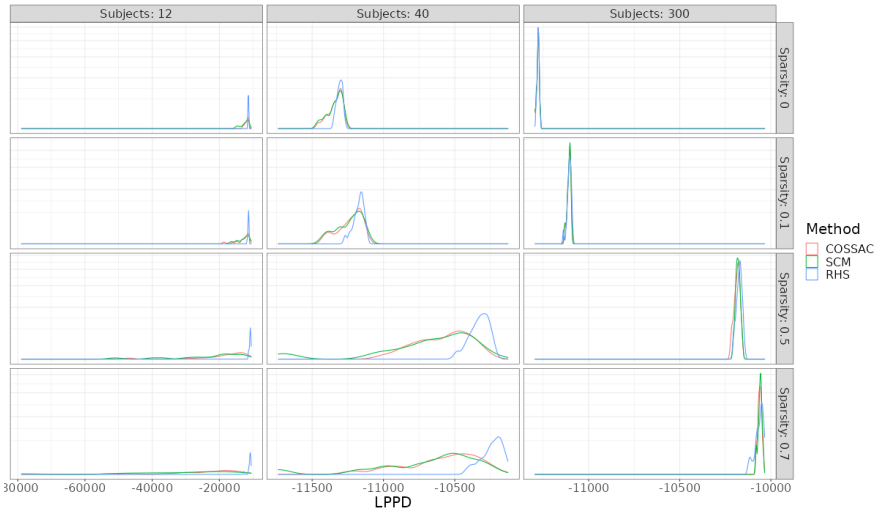


Figure S10 shows the actual density estimated across the 100 replicated datasets. The RHS shows not only a higher mean LPPD across replicated datasets, but also a much lower variance, showing that the regularization and Bayesian posterior averaging help to bring down the variance of the predictive error. Note also that LPPD goes up for higher sparsity settings and more covariates are able to help predict individual effects better.

Figure S11 shows the pointwise comparison of LPPD for RHS versus COSSAC. Again, the heavy tail of COSSAC is apparent, and for most individual datasets the RHS outperforms COSSAC.

Figure S12 explicitly shows the probability, estimated from the 100 replicates, of RHS outperforming each method respectively. Even at 300 subjects the RHS still outperforms on over 65% of datasets at all sparsity settings.

To provide another view of the improved predictive error on held-out data we also examine the log-residuals, $\epsilon = \log(y/\hat{y})$ where y is the held-out data from the 300 test subjects and $\hat{y}$ is the expectation of the predicted value from the model fit only on the subjects in the training data. For each replicated dataset we compute the standard deviation (SD) of these log-residuals. Figure S13 shows the SD of these log-residuals averaged over the 100 replicated datasets in each of the 12 simulation settings. The RHS shows a clear advantage in the variance of the held-out residuals that increases for the less sparse simulation settings and expectedly decreases as the number of subjects goes up.

Lastly, we provide a measure of predictive calibration of each method by considering the proportion of held-out data that falls within the 90% prediction interval. Specifically, for each observation in the test set, we compute a 90% prediction interval, again using only the model fit on the training data. We then compute the proportion of held-out data that fell within that interval for each

Figure S11: LPPD of RHS versus COSSAC for the 100 draws of each simulation setting.

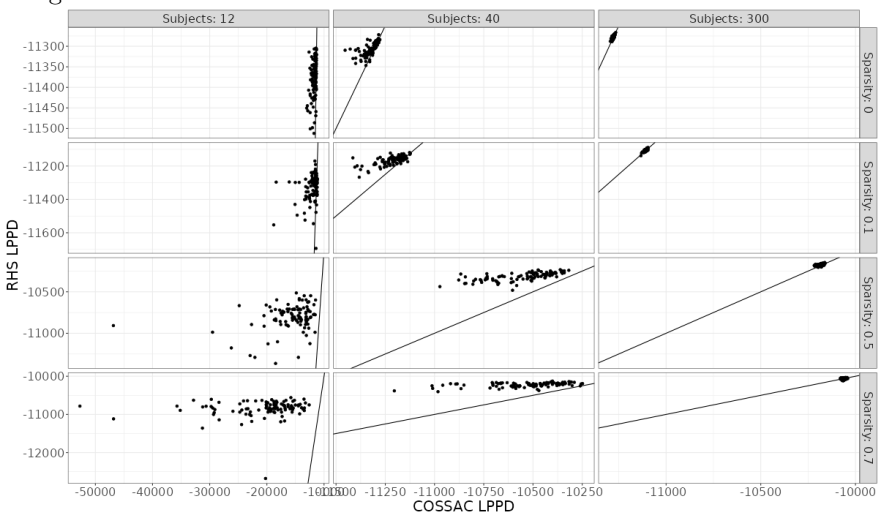


Figure S12: Estimated probabilities of RHS having a higher LPPD versus SCM and COSSAC respectively.

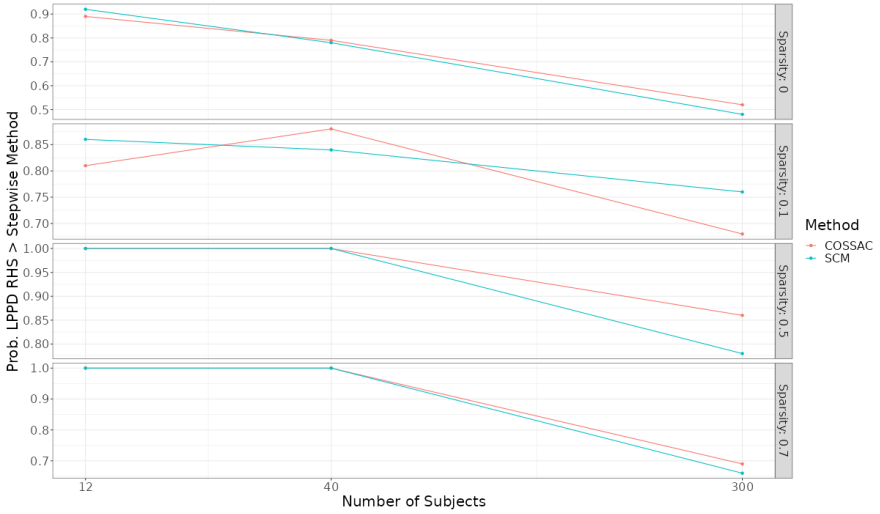
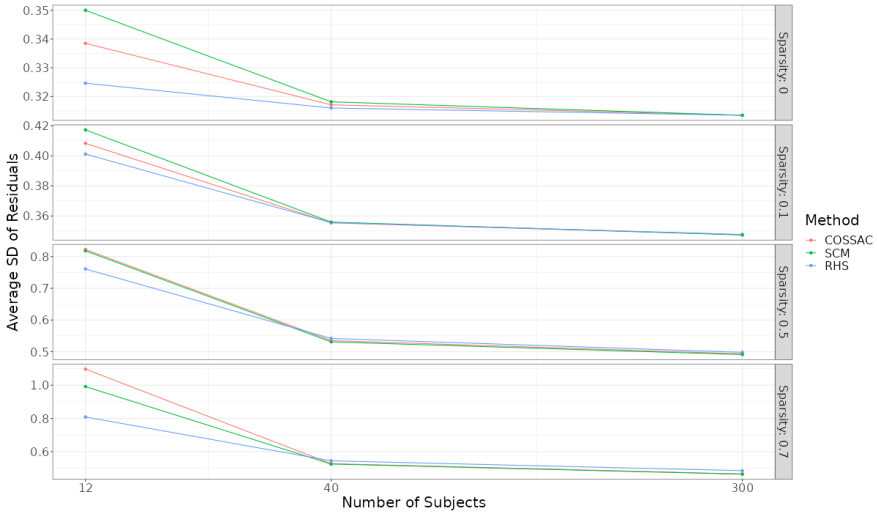


Figure S13: Average SD of log-residuals.



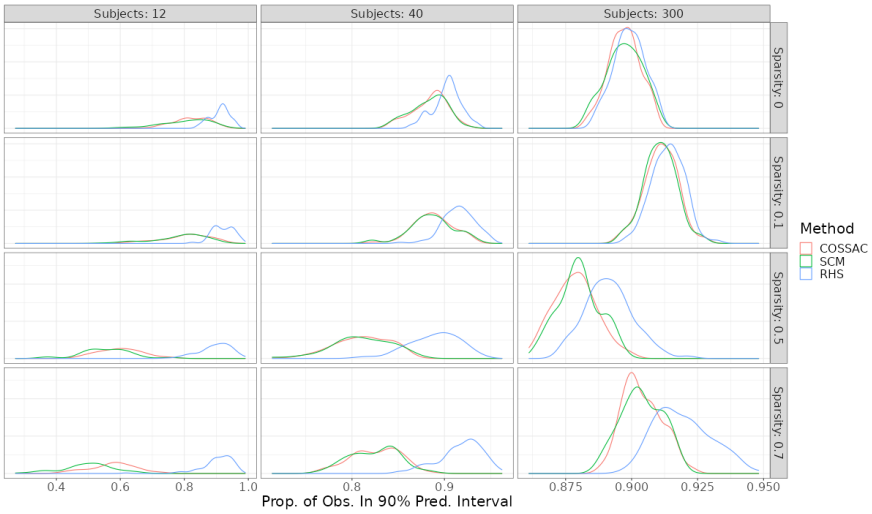
replication of each dataset in each simulation setting. Figure S14 shows densities of these computed proportions for each method in each simulation setting. For RHS the the computed proportions tend to be around the ideal value of 0.9 for all simulation settings. Meanwhile, the stepwise methods consistently show proportions less than 0.9 except for in the 300 subject setting. The miscalibration also seems to worsen for higher sparsity settings.

We note however that the RHS predictions inherently account for posterior uncertainty in the population parameters while the stepwise methods simply rely on point-estimates, so presumably the stepwise methods would show improved calibration if augmented with some bootstrapping scheme to account for parameter uncertainty. This of course would require orders of magnitude more runs of the stepwise methods which highlights an inherent benefit to the RHS method.

3 Prior Sensitivity Analysis and Alternate Posterior Summaries

Table S3 is a continuation of Table 2 and displays the LPPD of the RHS fit using alternative hyperparameters. One setting uses a sparsity prior that is too low $\tau_0 = 0.00126$ and gives prior predictive checks of 0%, 0%, and 6% sparsity values for the 5th, median, and 95th percentiles. The second setting is a prior that is too high $\tau_0 = 5$ with percentiles of 65%, 82%, and 92%. Even with these differently-specified priors, RHS at worst does as well as the other methods and usually still outperforms them. We do note a slightly improved value for the Warfarin example, although this can be attributed to the "true sparsity value"

Figure S14: Density of proportion of observations in each simulation setting that fall within the 90% prediction interval.



of that example being on the high end of the datasets thus the poorly specified prior which allows for many non-zero values on average can do slightly better. Indeed for that setting, we note that the model also "selects" the AGE-KIN relationship which was not selected by the standard hyperparameter setting. However, we expect the poorly specified setting to do worse on average than the standard hyperparameter setting, and that is what we see across the examples.

The table also includes LPPD values of the RHS when the posterior median and posterior mode of the population parameters are used rather than averaging LPPD over the posterior. We see similar behavior as when the posterior mean was used.

Table S3: Description of datasets/models along with LPPD for RHS models with alternate hyperparameters and alternate posterior summaries.

Dataset	**Model**	**Subj.**	**Pars.**	**Covs.**	**LPPD RHS Poorly specified high sparsity**	**LPPD RHS Poorly specified low sparsity**	**LPPD RHS Posterior Median**	**LPPD RHS Posterior Mode**
Remifentanil PK	3cmt IV	65	6	6	-3606	-3632	-3610	-3634
Theophylline PK	1cmt Depot	12	7	3	-183	-184	-193	-195
Tobramycin PK	1cmt Depot	97	2	4	-286	-285	-292	-295
Warfarin PK/PD	1cmt Depot w/ Lag + Indirect Emax response	32	8	3	-1097	-1104	-1140	-1214